P 10.5 Find $u_{3,0}(\rho)$ for the hydrogen $(Z=1)$.

$n=3$ $\ell=0$

Thus $\lambda = 1+\ell+n_r = 3$ in eq. 10.28

$$\Rightarrow \frac{c_{k+1}}{c_k} = \frac{k+1-3}{(k+1)(k+2)}$$

$$\begin{cases} \dfrac{c_1}{c_0} = \dfrac{-2}{2} = -1 & c_1 = -c_0 \\ \dfrac{c_2}{c_1} = \dfrac{-1}{2\cdot 3} = -\dfrac{1}{6} & \Rightarrow c_2 = -\dfrac{1}{6} c_1 = \dfrac{1}{6} c_0 \\ \dfrac{c_3}{c_2} = 0 & \end{cases}$$

$$F(\rho) = \sum_{k=0}^{\infty} c_k \rho^k = c_0 + c_1\rho + c_2\rho^2 + \overset{=0}{c_3}\rho^3 =$$

$$= c_0\left(1-\rho+\frac{1}{6}\rho^2\right)$$

$$u(\rho) = \rho^{\ell+1} e^{-\rho/2} F(\rho)$$

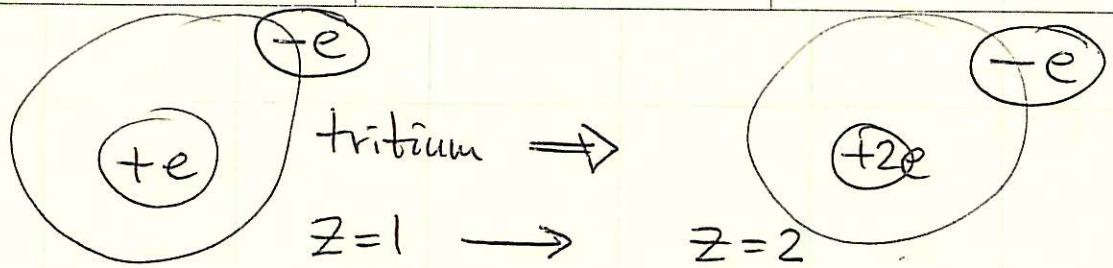
$$\Rightarrow u_{3,0}(\rho) = \rho e^{-\rho/2} c_0\left(1-\rho+\frac{\rho^2}{6}\right)$$

$$\rho = \frac{2Z}{n}\frac{r}{a_0} = \frac{2r}{3a_0}$$

$$\Rightarrow u_{3,0}(r) = c_0 \frac{2r}{3a_0}\left(1-\frac{2r}{3a_0}+\frac{2}{27}\frac{r^2}{a_0^2}\right) e^{-r/3a_0}$$

which agrees (up to undetermined c_0) with $R_{3,0}(r) = \dfrac{u_{3,0}(r)}{r}$ in eq. (10.45a)

P10.6



initial state

$$\psi_{in}(r) = R_{1,0}(r)\Big|_{Z=1} = 2\left(\frac{Z}{a_0}\right)^{3/2} e^{-Zr/a_0}$$

with $Z=1$

final state $\psi_{fin}(r) = R_{1,0}(r)\Big|_{Z=2}$

<u>Note:</u> in both cases the ground state is spherically symmetric $\Rightarrow$ we do need to consider the angular part of the wave function.

The amplitude to remain in the ground state:

$$\langle \psi_{fin} | \psi_{in} \rangle = \int d^3r \; R^*_{1,0}(r; Z=2)\, R_{1,0}(r; Z=1)$$

$$= \int_0^\infty dr \cdot r^2 \left(2 \frac{1}{a_0^{3/2}} e^{-r/a_0}\right)\left(2\left(\frac{2}{a_0}\right)^{3/2} e^{-2r/a_0}\right)$$

$$= \frac{4 \cdot 2^{3/2}}{a_0^3} \int_0^\infty dr\, r^2 e^{-3r/a_0} = \left(x = \frac{3r}{a_0},\; r = \frac{a_0 x}{3}\right)$$

$$= \frac{4 \cdot 2^{3/2}}{a_0^3}\left(\frac{a_0}{3}\right)^3 \underbrace{\int_0^\infty dx\, x^2 e^{-x}}_{2} = \frac{8 \cdot 2^{3/2}}{3^3} = 2^{3/2}\left(\frac{2}{3}\right)^3$$

$\Rightarrow$ The probability $= |\langle \psi_{fin} | \psi_{in} \rangle|^2 = \left(2^{3/2}\left(\frac{2}{3}\right)^3\right)^2 = 2^3\left(\frac{2}{3}\right)^6 =$

$$= \frac{2^9}{3^6} = \frac{512}{729} = 0.7$$

P10.7 Demonstrate the absence of solutions with $E < -V_0$ for the potential well $V = \begin{cases} -V_0, & r<a \\ 0, & r>a \end{cases}$

The radial equation

$$\left[-\frac{\hbar^2}{2\mu}\frac{d^2}{dr^2} + \frac{\ell(\ell+1)\hbar^2}{2\mu r^2} + V(r)\right] u(r) = E\,u(r)$$

$$\frac{d^2u}{dr^2} = \frac{2\mu}{\hbar^2}\left(\frac{\hbar^2\ell(\ell+1)}{2\mu r^2} + V(r) - E\right) u(r)$$

If $E < -V_0$ [or, in general, $E < \mathrm{Min}(V(r))$]

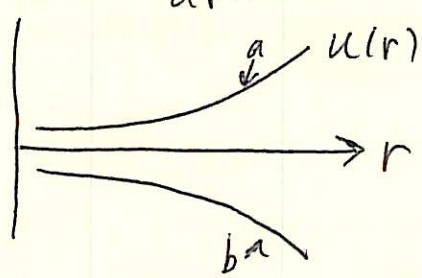
then the combination $A(r) \equiv \frac{\hbar^2\ell(\ell+1)}{2\mu r^2} + V(r) - E > 0$

is always positive!

$$\Rightarrow \frac{d^2u}{dr^2} = \frac{2\mu}{\hbar^2} A(r)\, u(r) \quad \text{with } \underline{A(r) > 0}$$

This means that $\frac{d^2u}{dr^2}$ and $u(r)$ always have the same <u>sign</u>

(a) Thus if $u(r) > 0$, then $\frac{d^2u}{dr^2} > 0$ everywhere, so that $u(r)$ is always increasing!



$\Rightarrow$ such function can not be normalized $\int_0^\infty dr\, |u(r)|^2 = \infty$

(b) If $u(r) < 0$, then $\frac{d^2u}{dr^2} < 0$, and $|u(r)|$ is increasing without restrictions again.

This is generally valid statement.

$\boxed{P10.10}$ $V(\rho) = \begin{cases} 0 & \rho < a \\ \infty & \rho > a \end{cases}$ $\rho = \sqrt{x^2+y^2}$

From <u>P9.22</u> $\nabla^2 \psi = \frac{1}{\rho}\frac{\partial}{\partial \rho}\left(\rho \frac{\partial \psi}{\partial \rho}\right) + \frac{1}{\rho^2}\frac{\partial^2 \psi}{\partial \phi^2} + \frac{\partial^2 \psi}{\partial z^2}$

Since $V(\rho) = \infty$ for $\rho > a$,

$\psi(r) = 0$ for $\rho > a$.

Consider now $\rho < a$:

$$-\frac{\hbar^2}{2\mu}\nabla^2\psi + \underset{\substack{\parallel \\ 0}}{V(r)}\psi = E\psi$$

$$\nabla^2\psi = -\frac{2\mu}{\hbar^2}E\psi$$

(a) $\psi(\rho, \phi, z) = R(\rho)\, Y(\phi)\, F(z)$ such that

$$\hat{L}_z\, Y(\phi) = m\hbar\, Y(\phi) \Rightarrow Y(\phi) = e^{im\phi}$$

$$\hat{p}_z\, F(z) = p_z\, F(z) \Rightarrow F(z) = e^{ip_z \cdot z/\hbar}$$

Also note that $\frac{\partial^2 \psi}{\partial \phi^2} = -\frac{1}{\hbar^2}\hat{L}_z^2\psi$

$$\frac{\partial^2\psi}{\partial z^2} = -\frac{1}{\hbar^2}\hat{p}_z^2\psi$$

$\Rightarrow$ if $\hat{L}_z\psi = \hat{p}_z\psi = 0$

$$\Rightarrow \frac{1}{\rho}\frac{d}{d\rho}\left(\rho\frac{dR}{d\rho}\right) + \frac{2\mu E}{\hbar^2}R = 0$$

Denote $k^2 = \frac{2\mu E}{\hbar^2}$

$$\frac{d^2R}{d\rho^2} + \frac{1}{\rho}\frac{dR}{d\rho} + k^2 R = 0 \quad (*)$$

Introduce $x = \rho k$, $\rho = \frac{x}{k}$

$$\Rightarrow \frac{d}{d\rho} = \frac{d}{dx}\frac{dx}{d\rho} = k\frac{d}{dx}$$

Then (*) becomes

$$k^2 \frac{d^2R}{dx^2} + k^2 \frac{1}{x} \frac{dR}{dx} + k^2 R = 0$$

Divide by k^2 and multiply by x^2:

$$x^2 \frac{d^2R}{dx^2} + x \frac{dR}{dx} + x^2 R = 0$$

This is diff. equation for the Bessel function $J_0(x)$

$$\Rightarrow \quad R(x) = J_0(x)$$

(the other possibility – $N_0(x)$ – is not normalizable)

Find energy eigenvalues from boundary condition:

$$R(\rho = a) = 0 \Rightarrow R(ak) = J_0(ak) = 0$$

$\Rightarrow$ need to know zeroes of $J_0(z) = 0$:

$$z_1 = 2.40 \; ; \; z_2 = 5.52 \; ; \; z_3 = 8.65$$

$$\Rightarrow \quad ka = z_n \quad \Rightarrow \quad \frac{2\mu E_n}{\hbar^2} = \frac{z_n^2}{a^2}$$

$$E_n = \frac{\hbar^2}{2\mu a^2} (z_n)^2$$

$$\Rightarrow \begin{cases} E_1 = \dfrac{\hbar^2}{2\mu a^2} (2.40)^2 \\ E_2 = \dfrac{\hbar^2}{2\mu a^2} (5.52)^2 \\ E_3 = \dfrac{\hbar^2}{2\mu a^2} (8.65)^2 \end{cases}$$

(b) Now $\hat{p}_z \psi = 0$ but $\hat{L}_z \psi = m\hbar \psi$

Then $$\nabla^2 \psi = \frac{\partial^2 \psi}{\partial \rho^2} + \frac{1}{\rho}\frac{\partial \psi}{\partial \rho} - \frac{\hat{L}_z^2 \psi}{\rho^2 \hbar^2} = -\frac{2\mu E}{\hbar^2}\psi$$

$$\Rightarrow \hat{L}_z^2 \psi = \hat{L}_z^2 R Y = R (m\hbar)^2 Y = m^2 \hbar^2 R Y$$

$$\Rightarrow \frac{d^2 R}{d\rho^2} + \frac{1}{\rho}\frac{dR}{d\rho} - \frac{m^2}{\rho^2} R = -k^2 R$$

Again $\rho = \frac{x}{k}$:

$$k^2 \frac{d^2 R}{dx^2} + k^2 \frac{1}{x}\frac{dR}{dx} - k^2 \frac{m^2}{x^2} R + k^2 R = 0$$

Thus $$x^2 \frac{d^2 R}{dx^2} + x \frac{dR}{dx} + (x^2 - m^2) R = 0$$

$$\Rightarrow R(x) = J_m(x)$$

Boundary condition: $R(ka) = 0 = J_m(ka)$

From the ~~book~~ webpage of <u>wolfram research</u> (to which the link is given on the homework) we observe that 3 lowest zeroes of $J_m(z)$ are provided by (in increasing value)

$m=0 \quad z_1 = 2.40$

$m=1 \quad z_2 = 3.83$

$m=2 \quad z_3 = 5.14$

$$\Rightarrow E_1 = E_{1,0} = \frac{\hbar^2}{2\mu a^2}(2.40)^2 \quad (m=0)$$

$$E_2 = E_{2,1} = \frac{\hbar^2}{2\mu a^2}(3.83)^2 \quad (m=1)$$

$$E_3 = E_{3,2} = \frac{\hbar^2}{2\mu a^2}(5.14)^2 \quad (m=2)$$